Working with Selenium-Firefox

Maven project to Connect all above 3 websites to Web Servers using Selenium Dependency on Firefox

Step 1: create a maven project on terminal by typing the command **mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenSeleniumApp05 -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false**

```
vboxuser@prathwish03:~/devops$ mvn archetype:generate \
  -DgroupId=com.example \
  -DartifactId=MyMavenSeleniumFirefox \
  -DarchetypeArtifactId=maven-archetype-quickstart \
  -DinteractiveMode=false
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< org.apache.maven:standalone-pom >-------------------
[INFO] Building Maven Stub Project (No POM) 1
[INFO] --------------------------------[ pom ]---------------------------------
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO]
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO] ----------------------------------------------------------------------------
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO] ----------------------------------------------------------------------------
[INFO] Parameter: basedir, Value: /home/vboxuser/devops
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MyMavenSeleniumFirefox
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/vboxuser/devops/MyMavenSeleniumFirefox
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  5.673 s
[INFO] Finished at: 2026-03-23T14:16:13+05:30
```

Step 2 : edit the pom.xml file and add the dependencies The pom.xml has the code

```
Open   pom.xml   ~/Devops/MyMavenSeleniumApp05   Save
<project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/maven-v4_0_0.xsd">
  <modelVersion>4.0.0</modelVersion>
  <groupId>com.example</groupId>
  <artifactId>MyMavenSeleniumApp05</artifactId>
  <packaging>jar</packaging>
  <version>1.0-SNAPSHOT</version>
  <name>MyMavenSeleniumApp05</name>
  <url>http://maven.apache.org</url>
  <dependencies>
    <dependency>
      <groupId>junit</groupId>
      <artifactId>junit</artifactId>
      <version>3.8.1</version>
      <scope>test</scope>
    </dependency>
    <dependency>
    <groupId>org.seleniumhq.selenium</groupId>
    <artifactId>selenium-java</artifactId>
    <version>4.41.0</version>
    <scope>compile</scope>
</dependency>
  </dependencies>
  <!-- Build: Configuring plugins and build settings -->
<build>
<plugins>
<!-- Example: Maven Compiler Plugin to compile Java code -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<version>3.8.1</version>
<configuration>
<source>1.7</source>
<target>1.7</target>
</configuration>
</plugin>
```

pom.mal
~/Devops/MyMavenSeleniumApp05

```xml
<plugin>
            <groupId>org.apache.maven.plugins</groupId>

            <artifactId>maven-jar-plugin</artifactId>
            <version>3.0.1</version>
           <configuration>
                <archive>
                    <manifestEntries>
                        <Main-Class>com.example.App</Main-Class>
                    </manifestEntries>
                </archive>
            </configuration>


        </plugin>
        <plugin>
      <groupId>org.apache.maven.plugins</groupId>
      <artifactId>maven-shade-plugin</artifactId>
      <version>3.5.0</version>
      <executions>
        <execution>
          <phase>package</phase>
          <goals>
            <goal>shade</goal>
          </goals>
          <configuration>
            <transformers>
              <transformer implementation="org.apache.maven.plugins.shade.resource.ManifestResourceTransformer">
                <mainClass>com.example.App</mainClass>
              </transformer>
            </transformers>
          </configuration>
        </execution>
      </executions>
    </plugin>

</plugins>
</build>
</project>
```

Step 3: Edit the App.java

*App.java
~/Devops/MyMavenSeleniumApp05/src/main/java/com/example

```java
package com.example;
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.WindowType;
import org.openqa.selenium.firefox.FirefoxDriver;
import org.openqa.selenium.firefox.FirefoxOptions;
import org.openqa.selenium.interactions.Actions;
import org.openqa.selenium.support.ui.ExpectedConditions;
import org.openqa.selenium.support.ui.WebDriverWait;
import java.time.Duration;
public class App {

    public static void main(String[] args) throws InterruptedException {

        FirefoxOptions options = new FirefoxOptions();

        WebDriver driver = new FirefoxDriver(options);

        driver.manage().window().setSize(new org.openqa.selenium.Dimension(1920,1080));

        WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(15));

        Actions actions = new Actions(driver);

        driver.get("https://www.saucedemo.com/");

        driver.findElement(By.id("user-name")).sendKeys("standard_user");

        driver.findElement(By.id("password")).sendKeys("secret_sauce");

        driver.findElement(By.id("login-button")).click();

        System.out.println("SauceDemo login successful");

        driver.switchTo().newWindow(WindowType.TAB);

        driver.get("https://automationexercise.com/products");
```

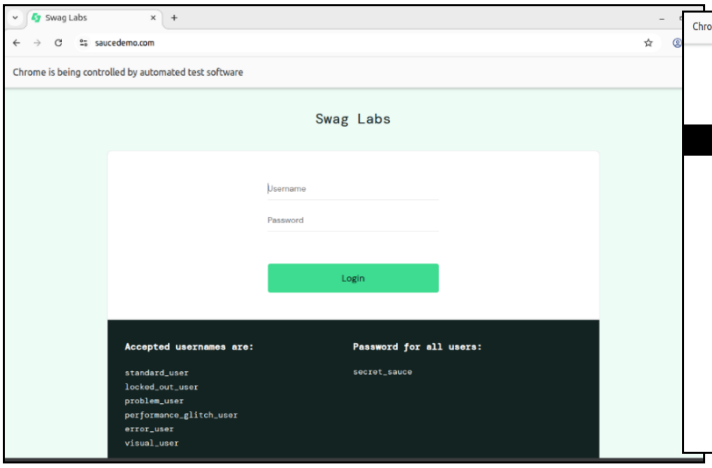
```
Open  *App.java  ~/Devops/MyMavenSeleniumApp05/src/main/java/com/example  Save
50          ((org.openqa.selenium.JavascriptExecutor) driver)
51                  .executeScript("arguments[0].scrollIntoView({block:'center'});", product);
52
53          ((org.openqa.selenium.JavascriptExecutor) driver)
54                  .executeScript("arguments[0].click();", product);
55
56          WebElement viewCart = wait.until(
57                  ExpectedConditions.elementToBeClickable(
58                          By.cssSelector("#cartModal a[href='/view_cart']")
59                  )
60          );
61
62          ((org.openqa.selenium.JavascriptExecutor) driver)
63                  .executeScript("arguments[0].scrollIntoView(true);", viewCart);
64
65          ((org.openqa.selenium.JavascriptExecutor) driver)
66                  .executeScript("arguments[0].click();", viewCart);
67
68          System.out.println("Automation Exercise product added to cart");
69
70          driver.switchTo().newWindow(WindowType.TAB);
71
72          driver.get("https://practicetestautomation.com/practice-test-login/");
73
74          Thread.sleep(2000);
75
76          driver.findElement(By.id("username")).sendKeys("student");
77
78          driver.findElement(By.id("password")).sendKeys("Password123");
79
80          driver.findElement(By.id("submit")).click();
81
82          System.out.println("Practice Test Automation login successful");
83
84          Thread.sleep(5000);
85
86          driver.quit();
87      }
88  }
```

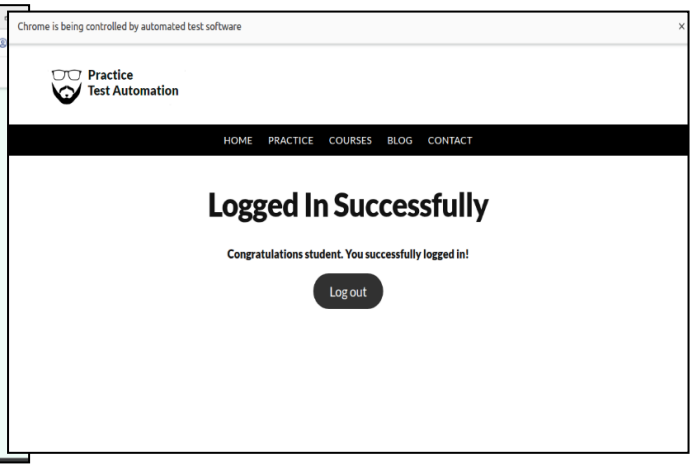
Step 4: Run mvn clean install

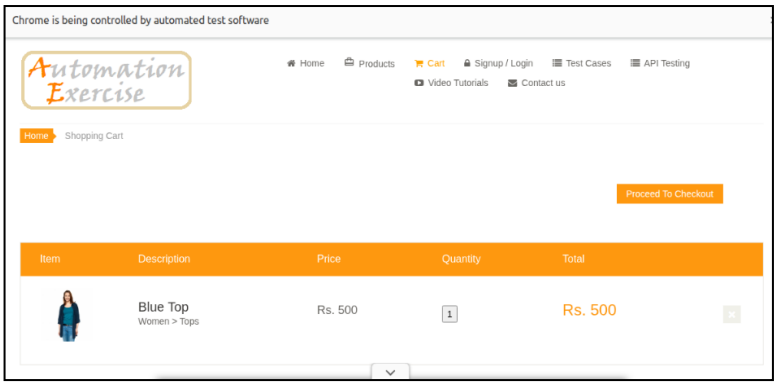
```
vboxuser@prathwish03:~/devops$ cd MyMavenSeleniumFirefox
vboxuser@prathwish03:~/devops/MyMavenSeleniumFirefox$ mvn clean install
[INFO] Scanning for projects...
[INFO]
[INFO] -----------------< com.example:MyMavenSeleniumApp01 >-----------------
[INFO] Building MyMavenSeleniumApp01 1.0-SNAPSHOT
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- maven-clean-plugin:2.5:clean (default-clean) @ MyMavenSeleniumApp01 ---
[INFO] Deleting /home/vboxuser/devops/MyMavenSeleniumFirefox/target
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ MyMavenSeleniumApp01 ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/vboxuser/devops/MyMavenSeleniumFirefox/src/main/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:compile (default-compile) @ MyMavenSeleniumApp01 ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file to /home/vboxuser/devops/MyMavenSeleniumFirefox/target/classes
[INFO]
[INFO] --- maven-resources-plugin:2.6:testResources (default-testResources) @ MyMavenSeleniumApp01 ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/vboxuser/devops/MyMavenSeleniumFirefox/src/test/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:testCompile (default-testCompile) @ MyMavenSeleniumApp01 ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file to /home/vboxuser/devops/MyMavenSeleniumFirefox/target/test-classes
[INFO]
[INFO] --- maven-surefire-plugin:2.22.2:test (default-test) @ MyMavenSeleniumApp01 ---
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
```

Step5:Build and run the maven application

To execute type the command **mvn exec:java -Dexec.mainClass="com.example.App"**it opens all the 3 websites automatically and registers using username and password.







MyMavenSeleniumFirefox pipeline

Step 1: Create new file named "**Jenkinsfile**" inside MyMavenSeleniumFirefox folder and push into git

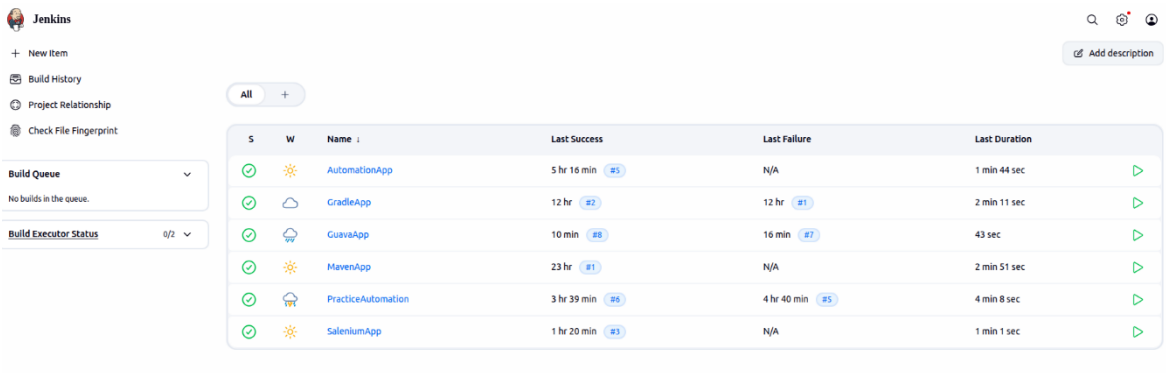
Jenkinsfile:

```
pipeline {
   agent any

   tools {
      maven 'Maven'
   }

   stages {

      stage('Checkout') {
         steps {
            git branch: 'main',
            url: 'https://github.com/PrathwishShetty03/MyMavenSeleniumFirefox.git'
         }
      }

      stage('Install Chrome & Dependencies') {
         steps {
            sh '''
            echo "Checking Chrome..."

            if ! command -v google-chrome > /dev/null
            then
               echo "Installing Chrome..."
               sudo apt update
               sudo apt install -y wget unzip

               wget https://dl.google.com/linux/direct/google-chrome-
stable_current_amd64.deb
               sudo apt install -y ./google-chrome-stable_current_amd64.deb
            fi

            echo "Chrome Version:"
            google-chrome --version
```

```
            echo "Installing required libraries..."
            sudo apt-get install -y libxss1 libappindicator1 libindicator7 fonts-liberation
libnss3 lsb-release xdg-utils
            '''
        }
    }

    stage('Build') {
        steps {
            sh 'mvn clean package -DskipTests'
        }
    }

    stage('Run Application') {
        steps {
            sh '''
            echo "Running using Maven..."
            mvn exec:java -Dexec.mainClass="com.example.App"
            '''
        }
    }
  }

  post {
    success {
        echo ' Build and execution SUCCESSFUL!'
    }
    failure {
        echo ' Build FAILED!'
    }
  }
}
```

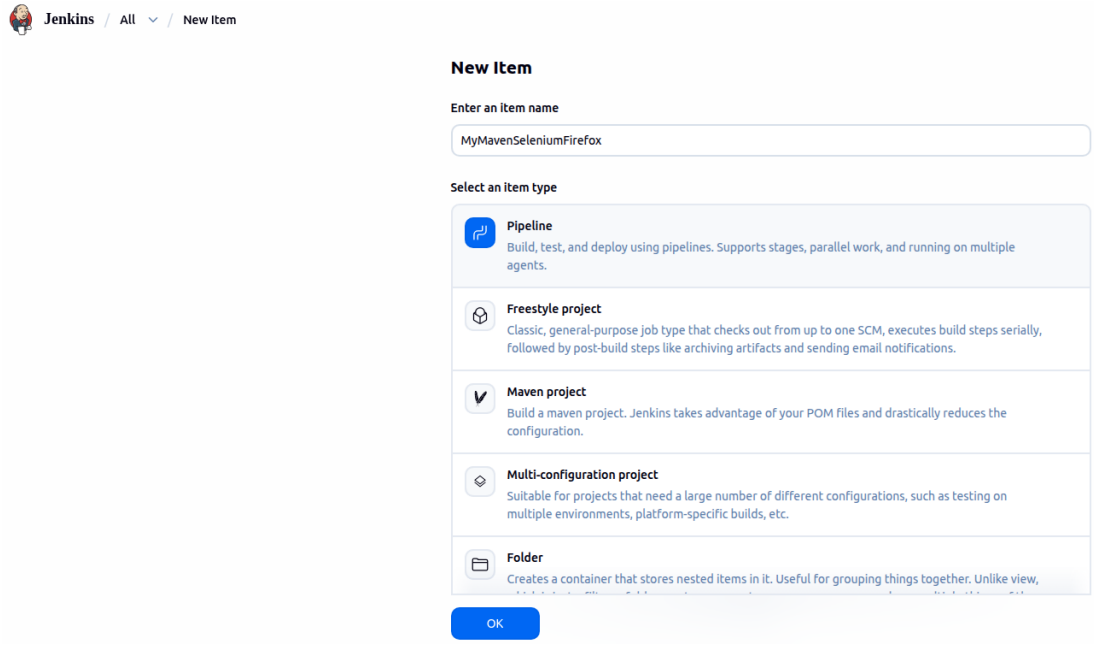
Step 2:Open Jenkins Dashboard

- Open Jenkins in browser
- Click on New Item



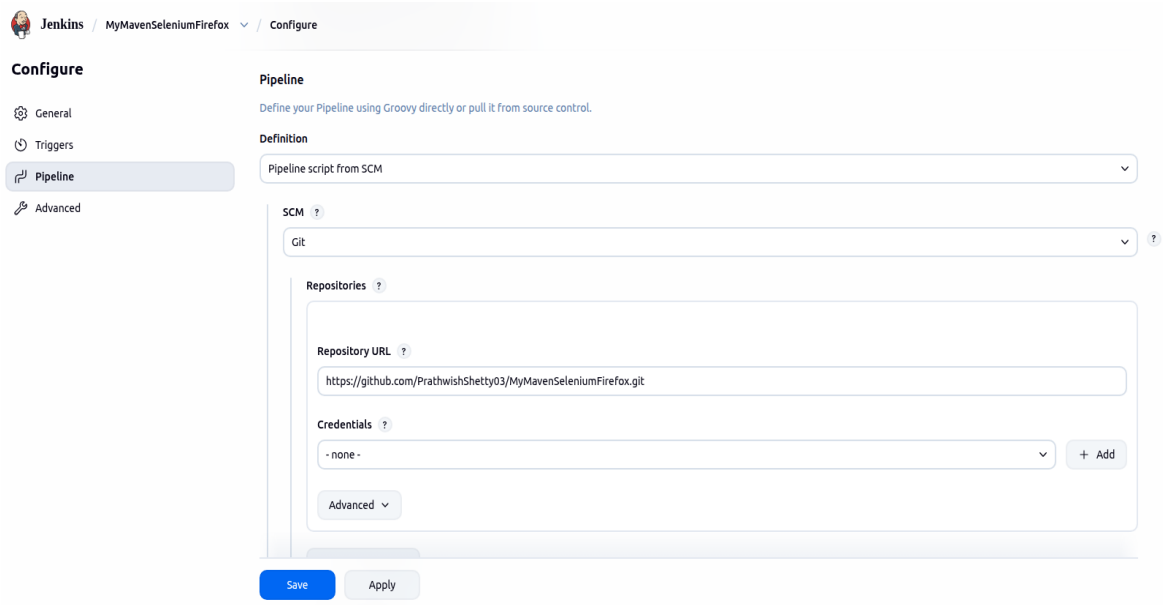
Step 3: Create Pipeline Job

- Enter name: MyMavenSeleniumFirefox
- Select **Pipeline**
- Click **OK**



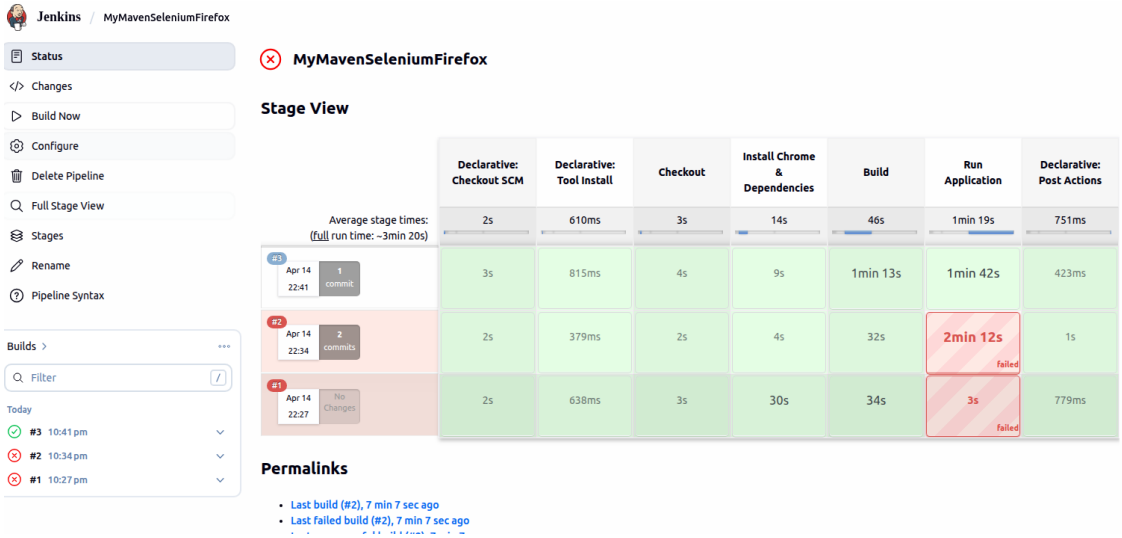
Step 4: Configure Pipeline with GitHub

- Select **Pipeline script from SCM**
- SCM → Git
- Enter repository URL
- Add credentials
- Click **Save**



Step 5: Run the Pipeline

- Click **Build Now**

Step 6: Look for the status of project

- Check Stage View
- View Build History